Feature Representation and Unsupervised Clustering: Comparing Raw Pixels and Histogram of Oriented Gradients on CIFAR-10

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Abstract

Feature representation of images are one of the most important ways to extract information from images to use for statistical modeling. Feature descriptors such as histogram of oriented gradients (HOG) provide a different representation of images than raw pixels. In supervised learning, HOG has established research and often outperforms raw pixel representations. However in unsupervised learning, HOG is less established. In this paper, we compare and evaluate raw pixels and HOG representations using K-means clustering under reduced dimensions using PCA. We show that through the silhouette scores from clustering that raw pixels provide more coherent and separable clusters than HOG through its difference in decomposition.

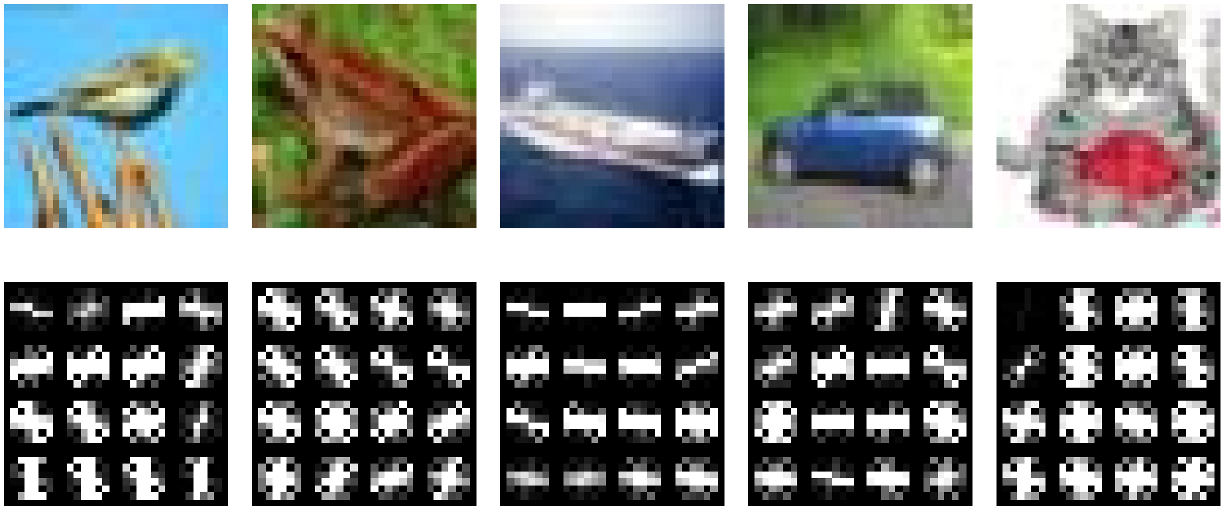

Figure 1: Sample 32x32 RGB images from CIFAR-10 and its corresponding histogram of oriented gradients (HOG)

1 Research Question

Does histogram of oriented gradients (HOG) offer more semantically meaningful representations of images instead of raw pixels through unsupervised clustering?

2 Background

In computer vision, images are seen as high-dimensional objects which are often hard to interpret when fed into statistical learning models due to its high dimension. To remedy this issue, other representations of images called feature descriptors are used as a substitute instead of raw pixels. A prominent example of a feature descriptor is the histogram of oriented gradients (HOG) [1]. HOG represents the image by patching the image into disjoint regions and per region, creates gradients from each pixel that point towards the direction of greatest pixel intensity change (See Figure 1).

Previous research on raw pixels and HOG for statistical modeling has been done through using supervised learning such as for face detection [2] and object detection [3] where HOG is shown to have more robust representations of the images. However, there has been no established research under the context of unsupervised learning on whether either representation has more coherent semantic meaning. To showcase this, we use CIFAR-10 [4], a dataset comprising of 60,000 32x32 pixel RGB images of 10 common objects to evaluate the differences between raw pixels and HOG representations.

3 Method

The test set of CIFAR-10 which consists of 10,000 images was chosen since it is a well-balanced representation of all the 10 categories in the dataset and to avoid computational complexities. For raw pixels, since the images are in RGB, it must first be converted from RGB to grayscale which is a weighted sum of the three channels into one channel. From there, the 32×32 images are flatten into vectors of $32 \times 32 = 1024$ dimensions. For HOG, we simply input the RGB images into the HOG algorithm as described in [1] using 12 patches and 8 orientations. Since clustering is prone to the curse of dimensionality which can severely impact on how metrics are measured [5], we use Principal Component Analysis (PCA) [6] in order to reduce the high-dimensional representations into lower dimensional representations using a small p amount of their principal components.

Given the p-dimensional PCA scores of each observation, we use K-means clustering [7] as the method for unsupervised learning to explore se-

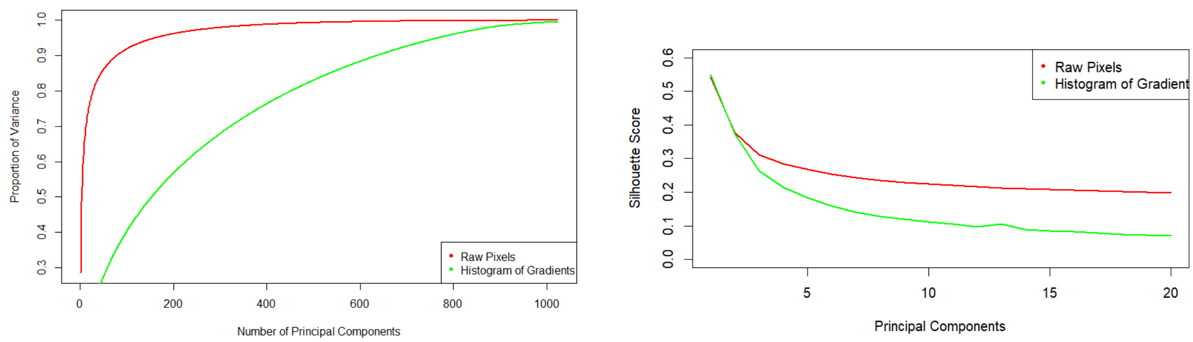

Figure 2: (Left) Proportion of variance explained by cumulative principal components. (Right) Number of principal components used and its corresponding cluster's mean silhouette score

mantically meaningful separations of the images in both representations. For both the raw pixels and HOG, we iterate for an increasing amount of p which will add more dimensional complexity. For each cluster, we use 10 different random initial centers to ensure the best clustering can be achieved from different trials. Each clustering with n observations is evaluated using the mean silhouette score [8] given as

$$\bar{s} = \frac{1}{n} \sum_i \frac{b_i - a_i}{\max(a_i, b_i)}$$

where a_i is the intra-cluster distance of observation i which is the average distance between observation i and other observations in the same cluster and b_i is the nearest-cluster distance of observation i which is the average distance between observation i and other observations from the nearest cluster.

We chose the mean silhouette score as the metric for clustering since it measures the quality of the cluster through its cohesion and separation. Cohesion is interpreted as how closely related a data point is to other data points within its own cluster. Separation is interpreted as how distinct a data point is from data points from its nearest neighboring cluster. In other words, it shows us how discernible and distinct each cluster is when distinct categories from CIFAR-10 also have distinct patterns from either representation. We use a grid search iteration approach to determine how many $k \in \{2, 3, ..., 10\}$ clusters provide the best silhouette score for each p.

To visually show the cohesion and separation of clusters, we use UMAP [9] which is a dimensional reduction technique for visualizing high-dimensional data that is still able to preserve local and global structure of data points and clusters. It reduces dimensions by constructing a weighted graph of the data and iteratively optimizes a matching low-dimensional layout that resembles the original high-dimensional graph as closely as possible. In this case, UMAP is used to reduce the p-dimensional clustering into a 2D scatterplot of it.

4 Results

Figure 2 shows the quantitative results of the different effects of increasing dimensional complexity by increasing the principal components used for both raw pixels and HOG representations. Both representations differ in the way the principal components cumulatively capture the variance of the images. While raw pixels exponentially capture the variances of the images, HOG much more slowly and approximately linearly capture the variances of the images. This shows that HOG has much less redundant information spread across its principal components since the first couple principal components are not able to capture a significant amount of the variance of the images unlike raw pixels as it is able to capture more than 90% using its first 100 principal components. Due to this, low-dimensional representations of HOG poorly capture information from the original image and gets increasingly difficult to capture the information due to the curse

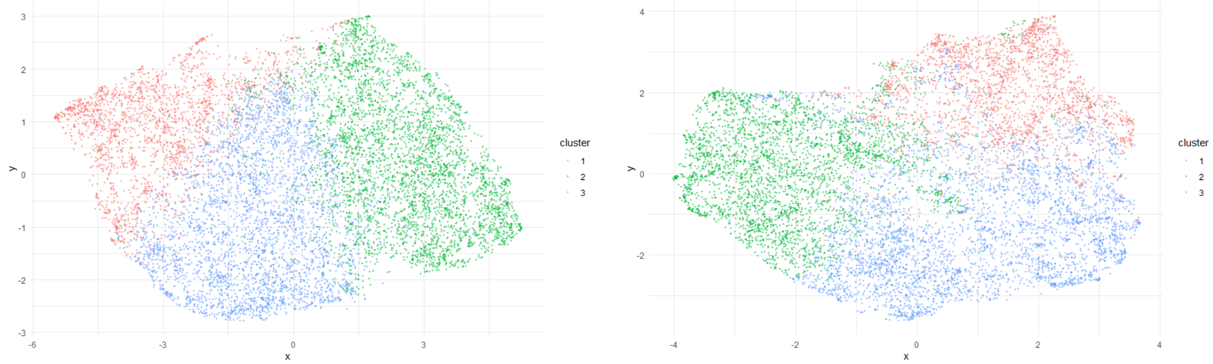

Figure 3: UMAP of K-means clustering at $p = 10$ principal components for raw pixels (left) and HOG (right)

of dimensionality as we add more dimensional complexity to the principal components.

For clustering a small dimensional representation of the raw pixels and HOG, there is a smooth decaying trend for the cluster's mean silhouette score as we add more dimensional complexity. While there is trivial difference for the mean silhouette score for both representations when using the first three principal components, this decay is worse for HOG as it nears the silhouette score of 0 meaning that the clusters aren't able to coherently capture any semantically coherent difference between each other. This aligns with the proportion of variance explained since HOG captures the information of the images significantly more slowly than raw pixels making the clusters from HOG's principal components less coherent. Additionally, increasing principal components increases the dimensional complexity of the clustering which causes HOG to not be able to use its high-dimensional more accurate representation of its principal components due to the curse of dimensionality. Hence, HOG has a bad representation for both low dimensions and high dimensions while raw pixels have a more stable representation in low dimensions explained by its better mean silhouette score than HOG.

Figure 3 shows the qualitative results of the clustering using UMAP. We chose $p = 10$ and $k = 3$ in order to show the effects of differing silhouette scores with low-dimensional p and small number of clusters p for visual clarity. Examining the boundaries of each cluster reveals that the raw pixels have better cohesion and separability since the clusters are more well-shaped and has less overlap among clusters creating clearer boundaries. Conversely for HOG, the boundaries are less coherent and separable since there is more overlap amongst clusters. These reflect the respective silhouette scores that the raw pixels and HOG achieved.

5 Discussion

In this paper, we used CIFAR-10 for unsupervised learning by using PCA and K-means clustering to distinguish whether raw pixel representations or HOG representations of images are more semantically meaningful. After clustering the low dimensional representations of both representations, we concluded that raw pixels offer a better representation of images compared to HOG due to better cohesion and separability of the clusters. While this research has been done on a dataset with varying image patterns, 32×32 pixel images are considered non-realistic low resolution images that do not reflect how real-world images work which may explain the lack of patterns that emerge from HOG. Furthermore, most of the best silhouette scores that came from the clustering is at $k = 2$ which can be hard to interpret.

Future work would include using more real-sized images that are of much higher resolution. Additionally, comparing to neural network activations would also be a natural comparison since feature descriptors are often used as preprocessing for neural networks.

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